

Benjamin Poole

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A [Google scholarship](#) recipient and researcher studying MSc in Artificial Intelligence. Graduated in Computer Science (1st Class) and was offered a funded PhD from my research experience.

EDUCATION

University of Edinburgh

MSc Artificial Intelligence

Edinburgh, UK

September 2026 – Present

University of Southampton

BEng Computer Science

Southampton, UK

September 2023 – July 2026

- Graduated with First Class Honours, achieving 89% in my machine learning module and completing my [dissertation](#) in agents and reinforcement learning.

RESEARCH EXPERIENCE

Researcher

March – September 2024

Agents Interaction and Complexity research group - University of Southampton

Researched the development of a vision language policy to control robot agents, supervised by a Postgraduate researcher.

- Developed a novel idea based on the MineCLIP dataset ([arxiv:2206.08853](#)) to build a teacher model that could train a student model using Imitation Learning.
- In my 2nd year I was invited back to research swarm robotics for [XPRIZE Wildfires](#) but the university's funding was cancelled. I have now been offered a funded PhD with this research group.

Researcher

October 2022 – August 2023

Tupel 3D

Researched and prototyped a low-cost, structured light hand-held 3D scanner.

- Self-taught computer vision and SLAM reconstruction from open online lectures.
- Developed a proprietary AI to create a 3D depth reconstruction system that could provide accurate 3D imagery using only primitive hardware. The company believe that this system is unique and superior to all publications they know.
- Developed a unique SLAM camera tracking system following ORB-SLAM 3 ([arxiv:2007.11898](#)) that would allow fast camera tracking for 3D reconstruction.

FEATURED PROJECTS

Projects demonstrated interactively on [my website](#).

From-scratch implementation and analysis of an LLM (2024)

[Write-up](#)

- Developed a 14.5m parameter language model from scratch (matrix multiplication), following the TinyStories paper ([arXiv:2305.07759](#)). The model was able to coherently model story language despite its small size.
- Analysed the model word vectors using 2D T-SNE dimensionality reduction to understand how the model interprets word semantics in terms of their context and relation to other words.
- Reading of academic papers to understand how LLM technologies work such as A Survey of Large Language Models ([arxiv:2303.18223](#)) and Locating and Editing Factual Associations in GPT ([arxiv:2202.05262](#)) which taught me LLM interpretability analysis and how LLMs store and retrieve facts.

Implementing a diffusion model to convert paint sketches to photorealistic images (2023) [Write-up](#)

- An implementation of the paper SDEdit ([arxiv:2108.01073](#)) to learn more about image prompting for generative AI.
- This involved experimenting with a diffusion model, following [arxiv:2112.10752](#) and then modifying an implementation of Stable Diffusion to add image prompting capability such that the mode could transform a simple MS Paint sketch to a photorealistic image.

Ferranti Mark 1 Assembler and Emulator (2025–26)

[Interactive demo](#)

- Built a browser-based emulator of the 1951 Ferranti Mark 1, with an assembler from a small assembly language to the machine's native code, including its five-bit teleprinter encoding, with live visualisation of the Williams-tube electronic store.

CONSULTING

Software Engineer

August – September 2025

Britwind Ecotricity

- Developed an IoT energy dashboard using web technologies, to display customer wind turbine power generation allowing full analytics and customisation in 6 weeks. This saved significant development costs by replacing a previously stalled, contractor-dependent effort.

OTHER INTERESTS

While my primary hobby is in computer science I also maintain an interest in electronics and philosophy.

- Twice winner of school science fair with audio electronics.
- Developed a solid-state Tesla Coil than can reach 60kV.
- ZVS (Zero Voltage Switching) induction heater from scratch.
- Built a 4 bit adder from 74xx series logic chips (with a friend) to demonstrate the workings of a CPU's ALU.
- I enjoy playing the drums and astrophotography: I have imaged the gravitationally lensed quasar (Q0957+561) 8.7 billion light years away using a DSLR camera with a tracking mount.